QML Lab • research project 12

QCNN for Phase Classification

Recognising the phases of the transverse-field Ising model directly from quantum states.

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Quadrant	QQ (quantum data, quantum processor)	**Track**	Quantum data
Difficulty	Hard	**Tools**	PennyLane, SciPy
Builds on	Modules 5 and 8 of the QML Lab	**Workspace**	`qml_projects/qcnn-phase-classification/`

Summary. A quantum convolutional neural network (QCNN) processes a many-body quantum state with layers of local unitaries and pooling, much like a convolutional network processes an image. Its input is the quantum state itself, which makes this the lab's main QQ project. The QCNN learns to tell the ordered from the disordered phase of the transverse-field Ising chain, and is compared with classical learners that only see measurement snapshots.

1 Background and key concepts

1.1 The transverse-field Ising model

The chain of N spins with Hamiltonian

$$H = -J \sum_{i=1}^{N-1} Z_i Z_{i+1} - h \sum_{i=1}^{N} X_i \tag{1}$$

has a quantum phase transition at $h = J$ in the thermodynamic limit (Pfeuty, 1970; Sachdev, 2011): for $h < J$ the ground state is ferromagnetically ordered, for $h > J$ it is paramagnetic. In finite chains the transition is smoothed and shifted, which a classifier must cope with.

1.2 Quantum convolutional neural networks

Cong et al. (2019) introduced the QCNN. A *convolution* layer applies the same two-qubit unitary to neighbouring pairs; a *pooling* layer measures (or traces out) half of the qubits and applies conditioned rotations to their neighbours; after $O(\log N)$ rounds a final unitary acts on the remaining qubit, whose $\langle Z \rangle$ gives the phase. The network has only $O(\log N)$ parameters. Pesah et al. (2021) proved that QCNNs do not suffer from barren plateaus, and Herrmann et al. (2022) ran a QCNN on a superconducting processor to recognise a symmetry-protected phase.

1.3 Quantum versus classical learners

A classical learner, as in Carrasquilla and Melko (2017), must work from measurement snapshots, losing phase information at every measurement. The QCNN acts on the state before measurement. Comparing both with the same number of state copies is the core of the QQ question.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 How close to the critical point $h = J$ does the QCNN classify correctly, and how does that change with N?

RQ2 Do QCNN gradients stay large as the chain grows, as theory predicts?

RQ3 How many state copies does the QCNN need compared with a classical model trained on measurement snapshots?

3 Method

3.1 Data

Ground states of the Ising chain for $N = 4$ and 8 (later 12) computed by exact diagonalisation with SciPy sparse methods, for h/J from 0.1 to 2.0. Training uses states away from the transition ($h < 0.8$ and $h > 1.2$); testing covers the full range.

3.2 Models

QCNN with parameterised two-qubit convolution blocks, pooling by controlled rotations and a final single-qubit readout; binary cross-entropy on the phase label; Adam. Input states are loaded exactly with `StatePrep` in simulation.

3.3 Baselines

Logistic regression on two-point correlations $\langle Z_i Z_j \rangle$ (an order-parameter-like feature); an MLP on computational-basis snapshots, as in Carrasquilla and Melko (2017); an unstructured hardware-efficient classifier with the same parameter count.

3.4 Metrics

- Accuracy far from and near the transition; predicted transition point h^* where $p = 0.5$.
- Gradient variance versus N.
- Number of state copies or shots to reach a given accuracy.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Small chain	$N = 4$	accuracy, h^*
E2	Larger chain	$N = 8$ (then 12)	accuracy, h^*
E3	Near criticality	test $h \in [0.8, 1.2]$	accuracy
E4	Copies	shots per state	accuracy vs snapshot MLP
E5	Trainability	QCNN vs unstructured ansatz	gradient variance

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/qcnn-phase-classification/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/qcnn-phase-classification/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "ising-phases"
title = "QCNN on transverse-field Ising ground states"
script = "experiments/e01_ising_phases.py"
figure = "phase_prediction.png"

[experiments.params]
n_spins = 8
h_min = 0.1
h_max = 2.0
n_train = 40
epochs = 100
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/qcnn-phase-classification
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_ising_phases.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Compare quantum and classical learners at an equal number of state copies.

7 Open directions

- The symmetry-protected topological phase of the cluster-Ising model, the original QCNN task.
- Inputs compressed by the quantum autoencoder (Project 06).
- Classical shadows as the classical learner's input (Project 13).

8 Risks and limitations

- In simulation the exact state is free; the cost of preparing it on hardware is the real bottleneck.
- Finite-size effects shift h^* away from 1; this is physics, not a bug.
- Training sets are small; regularise and use several seeds.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

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